

Activity 6: Root Cause with 5 Whys and Fishbone

Level: Intermediate Duration: 60 minutes Mapping: K7 A6 Tools: 5 Whys tool; Fishbone tool; evidence checklist

Goal

Use structured root-cause tools to separate evidence from assumptions and select a testable countermeasure.

Learner output

Problem statement, 5 Whys chain, fishbone map, evidence plan and countermeasure test.

Files in this folder

- README.md / README.pdf - activity guide and detailed procedure
- scenario.md / scenario.pdf - case facts and questions
- worksheet.md / worksheet.pdf - structured response template
- evidence-checklist.md / evidence-checklist.pdf - acceptance record
- data.csv - mock dataset when analysis is required

Detailed procedure

1. Write a measurable problem statement from the scenario evidence.
2. Open the 5 Whys tool and build more than one plausible causal branch.
3. Label statements as observed, calculated, reported or assumed.
4. Open the Fishbone tool and organise causes under People, Machine, Method, Material, Measurement and Environment.
5. Prioritise hypotheses by evidence, severity and ease of testing.
6. Define one disconfirming test for the leading hypothesis.
7. Use GenAI only to suggest overlooked questions; do not treat suggestions as evidence.
8. Select a countermeasure after the test and define the follow-up measure.

Verification

Complete every acceptance item in evidence-checklist.md. Submit the worksheet plus calculations, exported tool output or screenshots named in the worksheet.

Safety, privacy and responsible AI

Use only the supplied scenario data unless your organisation has authorised another dataset. Remove personal, confidential and export-controlled information. AI outputs are suggestions until verified by the named human decision owner.